

Agenda

1.) Announcements

- HW 11.8, 11.9, 11.10 due today
- Discussion Quiz 9 tomorrow over
- Midterm 2 Scores will be released today
- The drop deadline is Friday

2.) Finish 12.1

3.) Start 12.2

Equations in Three-Dimensions

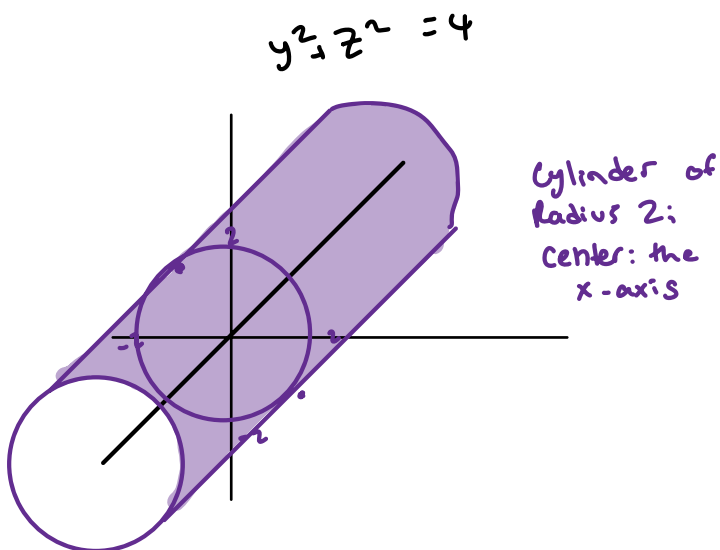
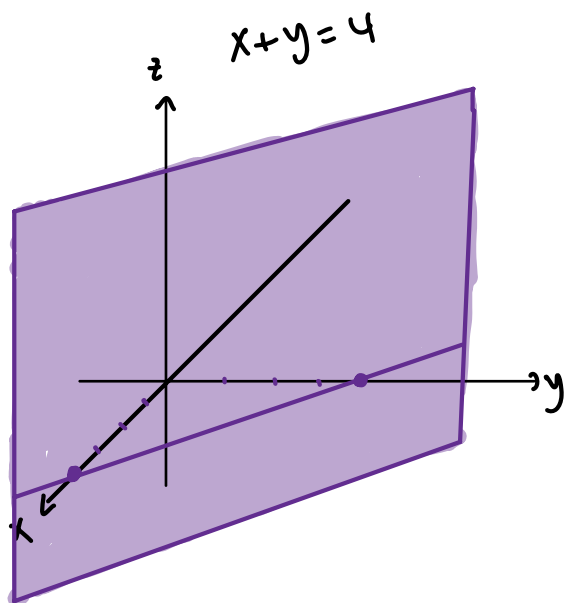
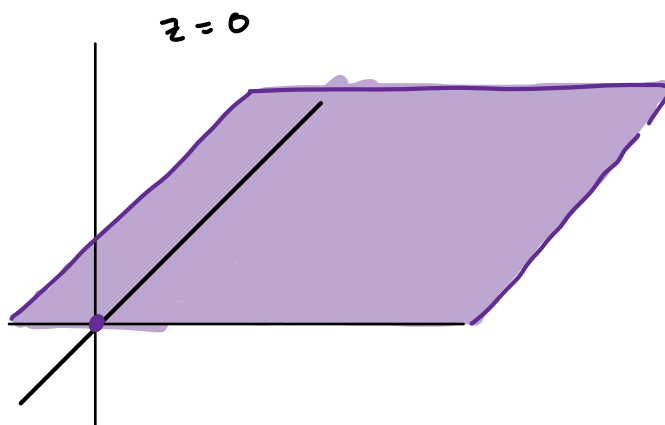
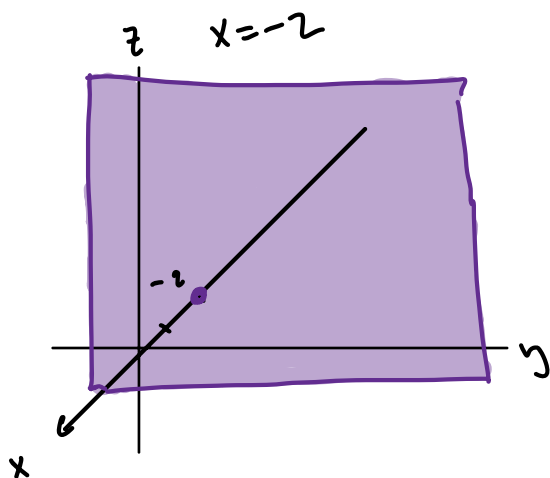
Example: Graph the following in three-dimensions:

1.) $x = -2$

2.) $z = 0$

5.) $x + y = 4$

6.) $y^2 + z^2 = 4$



Spheres

A sphere is a set of points with a fixed distance r , called the radius, from a particular point $(x, y, z) = (a, b, c)$, called the center.

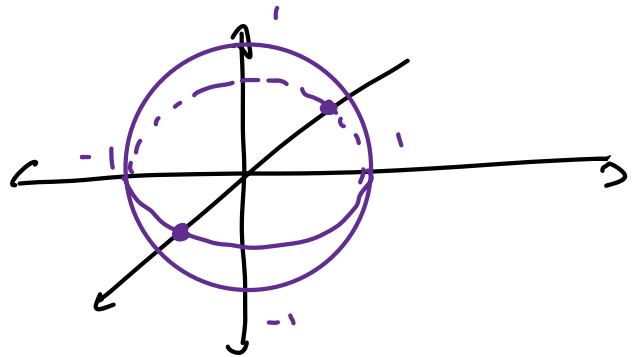
Let (a, b, c) be the center of a sphere with $r > 0$

For any point (x, y, z) on the sphere, the distance formula gives:

$$\sqrt{(x-a)^2 + (y-b)^2 + (z-c)^2} = r$$

Formula for sphere: $r^2 = (x-a)^2 + (y-b)^2 + (z-c)^2$

The unit sphere is centered at the origin with radius 1.



Example: Find the center and radius of the sphere $x^2 - 2x + y^2 + 4y + z^2 = 4$

$$(x^2 - 2x) + (y^2 + 4y) + z^2 = 4$$

$$(x^2 - 1) - 1 + (y + 2)^2 - 4 + (z - 0)^2 = 4$$

$$(x^2 - 1) + (y + 2)^2 + (z - 0)^2 = 9$$

$$\text{center} = (1, -2, 0); r = 3$$

$$(x^2 - 1)^2 = x^2 - 2x - 1$$

$$(x^2 - 1)^2 - 1 = x^2 - 2x$$

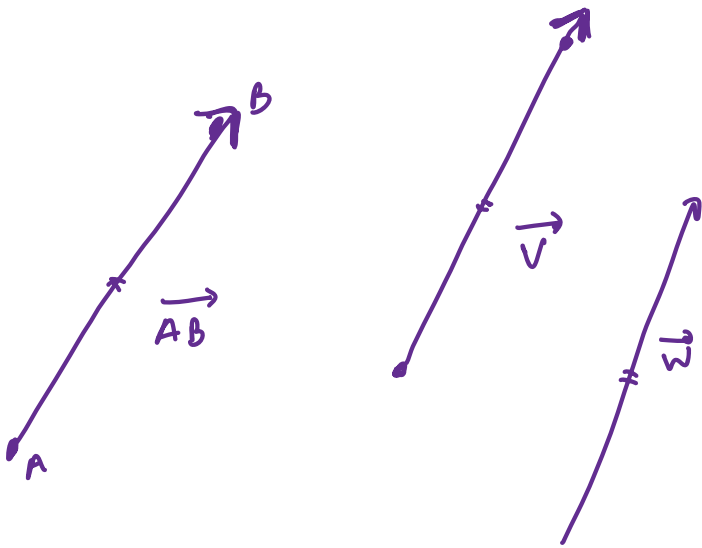
$$(y + 2)^2 = y^2 + 4y + 4$$

$$(y + 2)^2 - 4$$

2.2

Vectors

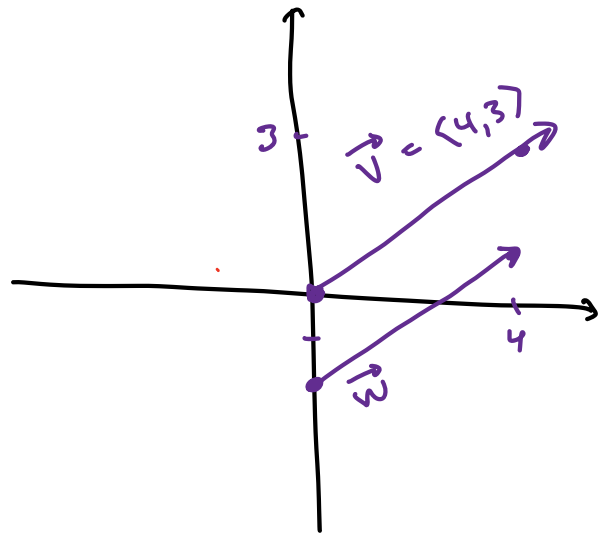
A vector is a coordinate symbolized by an initial point A and a terminal point B , with an arrow denoting direction



$$\overrightarrow{AB} = \vec{v} = \vec{w}$$

Equivalent vectors

- same length
- same direction



$$\vec{v} = \vec{w} = \langle 4, 3 \rangle$$

Let $\langle x_1, y_1, z_1 \rangle$ be an initial point and
 $\langle x_2, y_2, z_2 \rangle$ be a terminal point

then the vector has components

$$\langle x_2 - x_1, y_2 - y_1, z_2 - z_1 \rangle \text{ in } \mathbb{R}^3$$

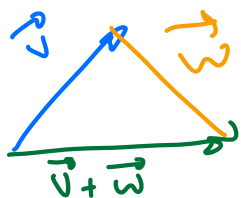
Components of Vectors:

Let $\vec{v} = \langle x_1, y_1, z_1 \rangle$ and $\vec{w} = \langle x_2, y_2, z_2 \rangle$. Let k be a constant.

Addition: $\vec{v} + \vec{w} = \langle x_1, y_1, z_1 \rangle$

$$+ \langle x_2, y_2, z_2 \rangle$$

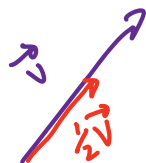
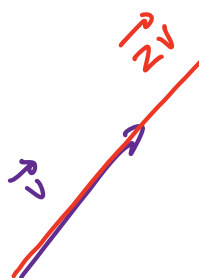
$$= \langle x_1 + x_2, y_1 + y_2, z_1 + z_2 \rangle$$



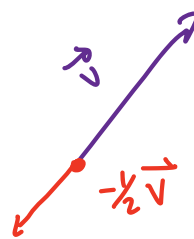
(k)

Scalar Multiplication: $k \cdot \vec{v} = k \langle x_1, y_1, z_1 \rangle$

$$= \langle kx_1, ky_1, kz_1 \rangle$$



$$0\vec{v} = \vec{0} = \langle 0, 0, 0 \rangle$$

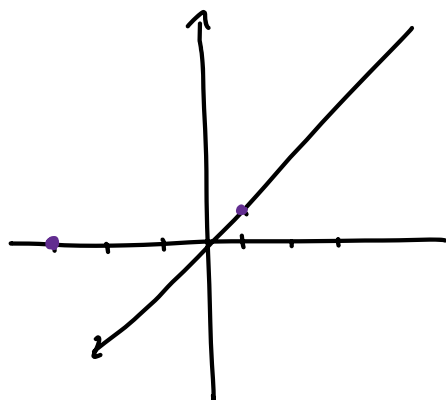
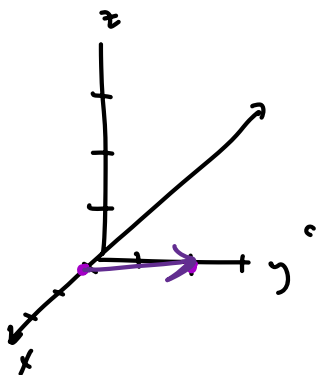


Activity: $\vec{v} = \langle 1, 2, 0 \rangle$, $\vec{w} = \langle -1, -3, 1 \rangle$. Find:

1.) $\vec{v} + \vec{w}$

2.) $-3\vec{v}$

3.) $-2\vec{v} + 5\vec{w}$



Parallelogram Law

For any vectors $\vec{v}$ and $\vec{w}$, we get $\vec{v} + \vec{w} = \vec{w} + \vec{v}$

The difference of two vectors $\vec{v}$ and $\vec{w}$ is defined as

$$\vec{v} - \vec{w} = \vec{v} + (-\vec{w})$$

Vector Length

Given $\vec{v} = \langle x, y, z \rangle$, its length or magnitude is given as

Activity: Find the indicated vector lengths:

1.) $|\langle 1, 2, 3 \rangle|$

2.) $|\langle \frac{3}{4}, -\frac{1}{4} \rangle|$